MIL-T-45312(Ord)
3 April 1959
SUPERSEDING
DAPD-176
2 August 1954

MILITARY SPECIFICATION

TANK RECOVERY VEHICLE: HEAVY, M51

1. SCOPE

1.1 Scope. - This specification covers an armored, full-tracked, rear-sprocket driven, heavy recovery vehicle.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

Federal

RR-R-571	-	Rope; Wire.
TT-E-485	-	Enamel, Semi-Gloss, Rust-Inhibiting.
TT-E-529	-	Enamel, Synthetic, Semi-Gloss.
VV-M-561	-	Gasoline, Automotive/Motor Fuel M.

Military

MIL-E-468	-	Extinguishers, Fire, Carbon Dioxide, (Portable, Permanent Shutoff, Hand and Wheeled Types).
MIL-L-644	-	Lubricating-Oil, Preservative, Special.
MIL-G-3056	-	Gasoline, Automotive, Combat.
MIL-L-3150	-	Lubricating-Oil, Preservative, Medium.
MIL-G-3278	-	Grease; Aircraft and Instruments (for Low and High Temperatures).
MIL-H-6083	-	Hydraulic Fluid, Petroleum Base, Preservative.
MIL-L-6085	-	Lubricating Oil; Aircraft Instrument, Low Volatility.
MIL-L-10295	-	Lubricating Oil, Internal-Combustion Engine, Sub-Zero.
MIL-S-10379	-	Suppression, Radio Interference, General Requirements for Vehicles (and Vehicular Sub-Assemblies).
MIL-G-10924	-	Grease, Automotive and Artillery.
MIL-B-11352	-	Block, Vision: Bullet Resistant.
MIL-S-12158	-	Sealing Compound, Noncuring, Polybutene.
MIL-P-13525	-	Plates; Designation, Identification, and Instruction, for Tank and Automotive Material.
MIL-H-13919	-	Hydraulic Fluid, Petroleum Base, Fire-Control.
MIL-G-14461	-	Quality Control; General Specification for.
MIL-C-16173	-	Corrosion Preventive, Solvent Cutback, Cold-Application.
MIL-L-21260	-	Lubricating Oil, Internal-Combustion Engine, Preservative.
MIL-T-45056	-	Tank Recovery Vehicle: Heavy, M51, Preparation for Storage and Shipment of.

FSC 2320

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MIL-E-45162 - Engine, Gasoline: 4-Cylinder, Horizontal Opposed, Fuel Injection, 127 H.P., 6-Cylinder, Horizontal Opposed, Fuel Injection, Air Cooled, 200 H.P., 375 H.P., and 500 H.P., and 12-Cylinder, 90-Degree V-Type, Fuel Injection, and Carbureted, Air-Cooled, 810-H.P. and 1,020 H.P.

STANDARDS

Federal

FEDERAL STD. NO. 595 - Colors.

Military

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-193 - Painting Procedures, Tracked Vehicles and Tactical Wheeled Vehicles.

MIL-STD-417 - Rubber Compositions, Vulcanized General Purpose, Solid (Symbols and Tests).

MIL-STD-642 - Identification Marking of Administrative, Combat, and Tactical Transport Vehicles.

DRAWINGS

Ordnance Corps

K8694420 - Diagram, Water Sealing.

Engineering Parts List

8358713 - Vehicle, Heavy Recovery, M51.

(Copies of specifications, standards, drawings, and publications, required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. - The following document forms a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply:

NATIONAL BUREAU OF STANDARDS

Handbook H28 - Screw-Threads Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D.C.)

3. REQUIREMENTS

3.1 Preproduction model. - When specified (see 6.2), a preproduction sample of the complete vehicle shall be submitted to the Government for approval in conformance with applicable drawings, specifications, requirements and tests specified herein.

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3.2 Materials. - Materials shall be as specified herein, and in the standards, specifications, and drawings listed or referred to in the Engineering Parts List (EPL) No. 8358713. In addition to requirements shown on the drawings, all rubber items which are under tension or flexed in any manner shall be ozone-proofed to meet the "C" test requirements of Standard MIL-STD-417.

3.3 Construction. - Vehicles, components, sub-assemblies and assemblies shall be fabricated and assembled into a complete vehicle in accordance with drawings listed or referred to in EPL 8358713.

3.3.1 Screw threads. - All screws, bolts, nuts, and studs shall be threaded in accordance with Handbook H28.

3.3.2 Sealing and seals. - Sealer conforming to type II of Specification MIL-S-12158 shall be used for sealing of vehicle in accordance with the water sealing diagram Drawing K8694420. The accumulation of water on the crew compartment hull floor shall be not more than 1 inch when vehicle has been standing in 60 inches of water for 30 minutes. Seals shall exclude water and other foreign material from all bearings to the extent specified in 3.5.7.

3.3.3 Winterization equipment. - When specified and installed (see 6.2), the winterization equipment shall assure satisfactory starting and operation in ambient temperature of minus 25° to minus 65°F.

3.3.4 Power plant. - The power plant consisting of engine, transmission center section, right and left reduction sections, cooling systems, and components and details specified on the engine and transmission assembly drawings shall be installed. The engine shall conform to type II of Specification MIL-E-45162.

3.3.5 Auxiliary generator. - The auxiliary generator consisting of an engine, generator and controls shall operate as starter generator combination. The generator shall have a constant output of 240 amperes, 28 volts, 6.7 kilowatt 3100 r.p.m. in ambient temperature of minus 65° to plus 125°F.

3.3.6 Fire extinguishing. - A permenently mounted 2-shot fire extinguishing system operable from inside and outside vehicle shall be installed. Two nonshatterable 5 pound capacity conforming to type I, and 2, 15 pound capacity fire extinguishers conforming to type III of Specification MIL-E-468, finished with semi-gloss enamel conforming to Specification TT-E-529 or type IV of Specification TT-E-485 color to match spot 24087 of Standard FEDERAL STD NO. 595 shall be installed.

3.3.7 Hydraulic oil reservoir. - The hydraulic reservoir shall evidence no leaking and shall be free of all foreign matter.

3.3.8 Hydraulic lines. - Prior to installation and final assembly all hydraulic lines shall be blown with dry air of a pressure that shall dislodge and expel all foreign matter.

3.3.9 Wire rope. - All wire rope used shall conform to Specification RR-R-571.

3.3.10 Stabilizing spades. - Each vehicle shall be equipped with a stabilizing spade at front and rear, operated by the auxiliary winch, with spades in operating position the vehicle shall be held stationary when winching maximum loads, and hoisting maximum loads, the vehicle shall not become embedded in the earth.

3.3.11 Controls. - Controls for the engine, transmission, electrical equipment, boom winches, permanently mounted fire-extinguishing system, armament and other operating mechanisms shall operate with ease and without interference.

3.3.12 Adjustments. - All adjusting mechanisms shall operate smoothly, without interference, and shall maintain adjustments under all operating conditions.

3.3.13 Fastenings and locks. - Fastening and locking devices shall be positive in action. Provisions shall be made to lock vehicle from outside of vehicle, using applicable locking devices.

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3.3.14 Fuel system. - The fuel system shall maintain a supply of fuel to the main and auxiliary engine when vehicle is ascending or descending a 60 percent grade or operating on 30 percent side slope.

3.3.14.1 Fuel tank. - The fuel tank shall be free of rust, scale, welding spatter and slag, and shall be protected inside and outside against corrosion.

3.3.14.2 Fuel lines. - Prior to installation, high pressure compressed dry air shall be forced through fuel lines to remove all foreign matter that may be lodged therein.

3.3.15 Communication system. - Provisions shall be made for installation of radio, interphone system and interphone extension cords. An interphone box shall be installed in the cab to facilitate the use of long leads for ground communications.

3.3.16 Vision blocks. - Vision blocks shall conform to Specification MIL-B-11352.

3.3.17 Stowed equipment. - On vehicle equipment (OVE), items shall stow (combat loaded) on the vehicle in spaces provided for the items so that items will not interfere with the use and operation of vehicle and components in any manner.

3.3.18 Repair welding. - Repair welding of any type or class shall be made only when, and to the extent specifically authorized by the procuring activity.

3.3.19 Standard lubricants. - Military standard lubricant products conforming to the following specifications should be used within the indicated types or grades specified in table I. Specific applications by grades or types for various expected temperature ranges shall be as specified in production lubrication charts or other applicable drawings.

Table I. - Table of specifications covering Military standard products

Fuels, lubricants and preservative materials	Specifications	Types or Grades
Corrosion preventives on lubricating	MIL-C-16173	Grade 1, 2, 3
Engine Oils:		
Operation and Preservative	MIL-L-21260	Grade 1, 2, 3
Subzero	MIL-L-10295	One Grade
Fuels:		
Gasoline, Combat	MIL-G-3056	Type I, II
Gasoline, Automotive	VV-M-561	Premium
Grease:		
Automotive and Artillery	MIL-G-10924	One Grade
Aircraft and Instrument	MIL-G-3278	One Grade
Hydraulic Oils:		
Fire Control	MIL-H-13919	One Grade
Preservative	MIL-H-6083	Type I
Lubricating Oils:		
Aircraft Instrument	MIL-L-6085	One Grade
Preservative, Medium	MIL-L-3150	One Grade
Preservative, Special	MIL-L-644	One Grade

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3.4 Break-in run.

3.4.1 Preparation. - Prior to break-in run, the vehicle shall be completely assembled and adjstments completed. Vehicle including all components requiring lubricating, shall be lubricated and serviced with fuels and fluids conforming to specifications specified in table I. The main engine governor shall be set to operate in conformance with Specification MIL-E-45162.

3.4.2 Oil pressure and temperature. - Prior to start of, or any time during break-in run after the engine has been stopped for at least 30 minutes, engine shall be restarted and operated at 800 to 1200 r.p.m. until lubricating oil is at operating pressure and temperature.

3.4.3 Distance operated. - Prior to tests, each vehicle shall be given a break-in run on smooth, level, hard-surfaced roads; except a section of the road shall be Government approved undulating (bump) road. The vehicle shall be operated as specified in table II and over the undulating section in a and b division of break-in run in both directions for the purpose of observing the function and performance of various parts, assemblies, boom locks, controls, and the suspension system.

Table II. - Speeds and distance for break-in run

Division of run	Speed m.p.h.	Distance miles
a	0 to 10	10
b	11 to 20	25
c	21 to maximum	5

3.4.4 Reverse operation. - After each division of the break-in run, vehicle shall be stopped, the engine allowed to idle for not less than 2 minutes, and the vehicle driven in reverse for a distance of not less than 50 feet.

3.4.5 Condition after run. - After completion of the break-in run, the engine shall idle smoothly. Prior to vehicle being submitted to the Government for acceptance all suspension wheel bearings shall be checked to determine if adjustments are correct and readjusted if necessary. There shall be no defects indicative of damaged components, or maladjustments that may cause faulty vehicle operation.

3.5 Performance. - A complete vehicle combat loaded (stowed) and serviced with Military standard products conforming to specifications specified in table I shall perform as specified herein. Ammunition may be simulated in size and weight. Vehicle serviced and equipped for existing climatic conditions shall operate satisfactorily in ambient temperature of minus 25° to plus 125°F. without external aids and down to minus 65°F. with external aids (see 3.3.3). Vehicle shall be operated on smooth, level, hard-surfaced roads; except a section of approved undulating (bump) road.

3.5.1 Speeds. - Vehicle shall operate at 10 percent of maximum speed in low gear for not less than 1 hour without damage to engine, transmission and components. Vehicle operating in high gear shall maintain a sustained speed of 20 m.p.h. with towed load of 60 tons.

3.5.2 Acceleration. - Vehicle shall accelerate from a standing start to 20 m.p.h. in high gear in not more than 12 seconds.

3.5.3 Climbing. - Vehicle shall ascend grades of 60 percent in forward and reverse gear without stalling or damage to power plant and power train.

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3.5.4 Operation on grades and side slopes. - Vehicle standing on a 60 percent grade for not less than 2 minutes with engine under no load at idle speed between (800 and 1200 r.p.m.), followed by engine inoperative for not less than 2 minutes, the engine shall restart in not more than 1 minute with vehicle headed up and down grade. Similar engine starting test shall be conducted on 30 percent side slopes first right side, then left side of vehicle up slope.

3.5.5 Braking.

3.5.5.1 Stopping. - Vehicle traveling at a speed of 20 m.p.h. shall stop within a distance of 45 feet from point of brake application.

3.5.5.2 Holding. - Vehicle standing on a 60 percent grade with service brakes applied for stopping shall hold vehicle stationary, and brakes locked for parking with all other holding devices inoperative, vehicle shall be held stationary.

3.5.6 Turning. - Vehicle shall turn to the right and left in full 360 degree pivot and to infinite in neutral steer.

3.5.7 Fording. - Vehicle shall ford level hard-bottom salt or fresh water 60 inches in depth including wave without special equipment. Vehicles standing in 60 inches of water for 30 minutes, not more than 1 inch water shall accumulate on the crew compartment hull floor. With deep water fording kit installed, vehicle shall ford level hard-bottom salt or fresh water up to the top of cab including bow wave. With vehicle standing in water up to top of cab for 30 minutes not more than 2 inches of water shall accumulate on the crew compartment floor. During the 30 minutes vehicle is standing in the 60 inches of water or in water up to top of cab, the main engine shall idle for 15 minutes, then be stopped for not less than 2 minutes. The engine shall restart in not more than 2 minutes with all accessories functioning satisfactorily during and after fording. After the shallow and deep water fording there shall be not more than 2 percent contamination of lubricants in transmission, engine, accessories, final drives, and the suspension system.

3.5.8 Trench crossing. - Vehicle shall cross trenches 36 inches in depth by 96 inches in width without stalling or damage to fenders.

3.5.9 Vertical obstacle. - Vehicle with front pintle removed shall cross vertical obstacle 36 inches in height, and with front pintle installed 31 inches in height.

3.5.10 Power train. - The power train shall operate throughout all gear and speed ranges and steering positions; without evidencing objectionable noises, loss of lubricants, brake grabbing or chatter when applied for stopping; or brake slippage when applied to hold vehicle stationary.

3.5.11 Cooling systems.

3.5.11.1 Engine cooling system. - The engine cooling system consists of 2 axial-flow fans with shrouds. Shrouds shall direct air to cool the engine and maintain cylinder head temperature of not more than 525°F. at spark plug gasket, with engine under full load in ambient temperature of 125°F.

3.5.11.2 Engine and transmission oil cooling system. - Engine and transmission oil cooling system of the oil-to-air heat-exchanger type shall maintain the engine oil temperature at cooler outlet at not more than 240°F. and the transmission oil temperature entering the cooler at not more than 300°F. when operated in ambient temperature of 125°F., under full load at .3 transmission maximum speed in all gear ranges.

3.5.12 Boom and hook. - Hydraulically operated crane boom shall lift loads specified in 3.5.12.1 at a minimum reach of 4 feet from the rear of the hull with vehicle on level ground and with spade emplaced. The boom shall include a telescoping extension reaching to a minimum of 8 feet from rear of hull, capable of

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lifting loads as specified in 3.5.12.1. The vehicle shall be provided with a boom elevation limiting valve which shall limit the maximum boom elevation. The boom shall traverse a minimum of 30 degrees the right and left of the center line of vehicle. The boom shall traverse a minimum of 60 degrees from left to right or from right to left, without load, in not more than 70 seconds. The boom retracted shall raise from the travel lock position to maximum elevation (hook height of 12 feet) without load in not more than 60 seconds. The boom hook shall raise from 1 foot above ground to 15 feet (boom fully raised and extended) without load in not more than 60 seconds.

3.5.12.1 Boom and hook capacities. - With boom extended and fully elevated, the hydraulic system shall raise the hook with a 15-ton load to within 6 inches of the hook block stop and the operation of the hook hoist winch hydraulic relief valve shall prevent lifting the hook with a 17.8-ton load to within 6 inches of the hook block stop. With boom retracted and fully elevated, the hydraulic system shall raise the hook with a 30-ton load to within 6 inches of the hook block stop and the operation of the hook hoist winch hydraulic relief valve shall prevent lifting the hook with 34-ton load to within 6 inches of the hook block stop. The hydraulic system shall lift the boom, when retracted, with a 30-ton load from travel lock position to full elevation, and the operation of the hydraulic relief valve of the boom raising winch shall prevent lifting of a 34-ton load to full elevation.

3.5.12.2 Winches main and auxiliary. - Two hydraulically operated winches, with mountings, hydraulic motors, lines, and controls shall be installed. Winches shall be operable either to the front or to the rear of the vehicle.

3.5.12.3 Main winch. - The main winch shall have a cable capacity of 300 feet of 1-1/4 inch diameter cable. It shall produce a line pull of not less than 72,000 lbs. in low gear on second wrap and the hydraulic relief valve shall prevent the exertion of a line pull of 74,500 lbs. or more. The winch, under a load sufficient to obtain proper spooling, shall reel in the cable on a bare drum at a rate of not less than 10 feet in 30 seconds.

3.5.12.4 Auxiliary winch. - The auxiliary winch shall have a cable capacity of 475 feet of 1/2-inch diameter cable. It shall produce a line pull of not less than 10,000 lbs. in low gear on bare drum (1/2 to 2/3 covered) and the hydraulic relief valve shall prevent the exertion of a line pull of 11,000 lbs., or more. The winch, under a load sufficient to obtain proper spooling, shall reel in the cable on a bare drum a rate of not less than 15 feet in 30 seconds.

3.5.12.5 Winch brakes. - Winch brakes, capable of holding the full load on the winches and requiring both manual application and release, shall be provided on the main and auxiliary winches. The main winch brake shall be of the automatic type which allows the brake to be applied and the load to be winched in. The brake shall then automatically apply and hold the load and prevent the line from unwinding after hydraulic power has been removed. The brake shall require manual release to allow spooling out.

3.5.12.6 Auxiliary hydraulic system power unit. - An auxiliary hydraulic power unit, with capacity to raise and traverse the boom, when main engine is inoperative, shall be installed. The unit shall be inoperable when the main engine is running. The unit shall provide a line pull on bare drum (1/2 to 2/3 covered) of the auxiliary winch of not less than 4,100 lbs. The hydraulic relief valve shall operate to prevent the exertion of a line pull of 4,700 lbs. or more.

3.5.13 Radio noise suppression. - Vehicle shall be radio interference suppressed in conformance with applicable requirements of Specification MIL-S-10379.

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3.6 Painting, marking and data plates.

3.6.1 Painting. - The exterior and interior of vehicle, components, subassemblies, assemblies, and parts that require painting shall be prepared and painted in accordance with Standard MIL-STD-193. Exterior of vehicle color and nonskid paint where required shall be olive drab matching color chip 24087 of FEDERAL STD NO. 595 (see 6.3).

3.6.2 Marking. - Vehicle marking shall be in accordance with Standard MIL-STD-642.

3.6.3 Identification and data plates. - Identification and data plates shall conform to Specification MIL-P-13525.

4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor's inspection. - Unless otherwise specified herein, the supplier is responsible for the performance of all inspection requirements prior to submission for Government inspection and acceptance. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order.

4.1.1 Quality control requirements. - The contractor shall provide and maintain an inspection and quality control system in accordance with Specification MIL-G-14461.

4.2 Government verification. - Quality assurance operations performed by the contractor will be subject to Government verification at unscheduled intervals. Verifications will consist of (a) surveillance of the operations to determine that practices, methods, and procedures of the inspection plan are being properly applied, and (b) Government product inspection to measure quality of product offered for acceptance. Failure of the contractor to promptly correct deficiencies discovered shall be cause for suspension of acceptance until correction has been made or until conformance of product to prescribed criteria has been demonstrated.

4.3 Preproduction sample. - Preproduction sample combat loaded (stowed) shall be submitted for inspection and tests as specified in 3.1. Inspection shall be performed at the place of manufacture. Failure of the preproduction sample to comply with any of the requirements specified, shall be cause for refusal of the Government to conduct a retest until corrective measures satisfactory to the Government have been taken.

4.4 Sampling.

4.4.1 Lot size. - A lot shall consist of all vehicles of one type from an identifiable production period submitted at one time for inspection.

4.4.2 Sampling for inspection. - For the purpose of visual, dimensional and primary functional inspection, a representative sample shall be selected from each lot in accordance with inspection level II of Standard MIL-STD-105, using the following table of classification of defects and acceptable quality levels:

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DEF. NO.	CHARACTERISTICS	MAJOR A.Q.L. 15 DEFECTS/100 UNITS	METHOD OF INSPECTION
1.	Engine	Malfunction, Unusual Noise, Leaks, Mounting Improper, Damage	Visual & Functional
2.	Power Train	Malfunction, Unusual Noise, Leaks, Mounting Improper, Damage	Visual & Functional
3.	Fuel System Components	Malfunction, Damage, Leaks	Visual & Functional
4.	Fuel & Oil Lines, Valves & Hose	Defective, Clearance Improper, Malfunction, Leaks	Visual & Functional
5.	Primer System Components	Pump Malfunction, Leaks	Visual & Functional
6.	Exhaust System Components	Damage, Leaks	Visual & Functional
7.	Cooling System Components	Malfunction, Damage, Leaks, Unusual Noise	Visual & Functional
8.	Electrical System Components (Includes wiring & Instrumentation	Malfunction, Damage, Unusual Noise	Visual & Functional
9.	Suspension System Components	Damage, Tire or Insert Separation Improper Clearance or Alinement, Welding Defects, Leaks, Improper Assembly, Installation Bolt or Nut Torque & Track Tension	Visual & Functional; Torque
10.	Hull: Upper & Lower	Structure & Welding Defects	Visual
11.	Controls & Adjusting Mechanisms	Malfunction, Improper Clearance	Visual & Functional
12.	Doors, Fenders, Grilles, Panels, Ducts, Stowage Brackets & Stops	Malfunction: Torsion Spring & Hold-Open Devices	Visual & Functional
13.	Drain valves	Valve Seating Improper	Visual

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14.	Cushions, Seats, Rests, & Crash Pads	Damage	Visual
15.	Seals	Damage, Leakage, Contamination of Lubricants	Visual
16.	Boom	Malfunction, Damage, Unusual Noise	Visual & Functional
17.	Winches	Malfunction, Damage, Unusual Noise	Visual & Functional
18.	Auxiliary Hydraulic System	Malfunction, Damage, Unusual Noise	Visual & Functional
19.	Auxiliary Generator Set	Malfunction, Damage, Unusual Noise	Visual & Functional
20.	Fire Extinguishing System	Incorrect Installation	Visual & Functional
21.	Braking Ability (Parking & Service)	Stopping Distance or Grade Holding Non-conformance, Equalization Improper, Unusual Noise	Functional
22.	Turning & Steering Ability	Non-conformance, Wander, Side Pull, Unusual Noise	Functional
23.	Speed	Non-conformance	Functional
24.	Acceleration	Non-conformance	Functional
25.	Fording	Excessive Leakage	Visual
26.	Workmanship	Improper fastening of nuts, screws, bolts, washers, cotter pins, safety wires, chain and lock pins, wiring clips, ground straps, grommets, grease fittings, stripped threads, and loose items, Boom and Cables Damage, Sharp Bends	Visual

DEF. NO.		MINOR A.Q.L. 100 DEFECTS/100 UNITS	METHOD OF INSPECTION
101.	Engine	Improper Component Assembly or Installation	Visual
102.	Power Train	Improper Component Assembly or Installation	Visual

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103.	Fuel System Components	Improper Assembly or Installation	Visual
104.	Fuel & Oil Lines, Valves & Hose	Improper Assembly or Installation	Visual
105.	Primer System Components	Improper Installation or Assembly	Visual
106.	D/Water Fording System Components	Control Malfunction, Improper Installation, Assembly or Adjustment	Visual & Functional
107.	Exhaust System Components	Improper Assembly, Installation or Clearance	Visual
108.	Cooling System Components	Improper Assembly or Installation	Visual
109.	Electrical System Components (Includes Wiring & Instrumentation)	Improper Assembly, Installation Circuit Identification, Clearance or Protection	Visual
110.	Hull: Upper & Lower	Support & Bracket Bolts Missing, Insecure	Visual
111.	Controls & Adjusting Mechanism	Improper Assembly or Installation Welding Defects	Visual
112.	Doors, Fenders, Grilles, Panels Ducts, Stowage Brackets & Stops	Improper Fits, Adjustments, Assembly or Installation, Defective Weldments, Seals or Hardware, Sheet Metal Defects	Visual & Functional
113.	Drain Valves	Improper Installation, Assembly or Control Adjustment	Visual & Functional
114.	Periscope Housing	Improper Assembly, Installtion, Door Fit or Tension, Welding Defects	Visual & Functional
115.	Cushions, Seats, Rests & Crash Pads	Improper Fit, Installation or Adhesion	Visual
116.	Boom	Improper Control, Adjustment	Functional
117.	Winches	Improper Control Adjustment	Functional

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118.	Paint, Protection & Anti-Skid Coatings	Application, Coverage & Materials Improper	Visual
119.	Lube Fittings & Lubrication	Fittings Missing, Defective or Improperly Installed, Lubrication Improper	Visual
120.	Decals, Markings, Data & Instruction Plates	Incomplete Data, Missing, Improper Location or Size	Visual
121.	Workmanship	Loose nuts and screws, bolts, washers, cotter pins, wiring slips, ground straps, grommets, grease fittings, stripped threads, damaged defective, improperly installed, binding, and interference, Boom cables improperly installed	Visual

4.5 Tests.

4.5.1 Performance. - After specified break-in run (see 3.4), vehicles shall be subjected to road and other performance tests as specified in table III. Vehicles shall be serviced with fuel, lubricants and fluids as specified in table I, but shall not be combat loaded for tests at place of manufacture. Prior to actual tests, engines shall be operated between 800 to 1200 r.p.m. to warm engine and lubricants to operating temperature. Operation of vehicles, unless otherwise indicated, shall be on smooth, level, hard-surfaced roads; except a section shall be of an approved undulating (bump) condition. Vehicle tests shall be conducted at location or locations shown in table III.

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Table III. - Location for tests

Title	Paragraph Reference Numbers	Place of Manufacture	Government Proving Ground
Road test - 8 mile	4.5.1.1	X	-
Speed test	4.5.1.2	X	X
Acceleration test	4.5.1.3	X	X
Climbing test	4.5.1.4	-	X
Engine starting test	4.5.1.5	X	X
Stopping test (Brakes)	4.5.1.6	X	X
Holding test (Parking brakes)	4.5.1.7	X	X
Turning test	4.5.1.8	X	X
Fording test (shallow water)	4.5.1.9	X (1)	X
Fording test (deep water)	4.5.1.9	-	X
Trench crossing test	4.5.1.10	-	X
Vertical-obstacle test	4.5.1.11	-	X
Power train test	4.5.1.12	X	X
Cooling system test	4.5.1.13	X (2)	X
Boom and hook test	4.5.1.14	X	X
Boom and hook capacity test	4.5.1.15	X	X
Winch test	4.5.1.16	X	X
Winch brake test	4.5.1.17	X	X
Auxiliary hydraulic system power unit test	4.5.1.18	X	X
Winterization equipment test	4.4.2		X
Stowed equipment test	4.4.3		X
Radio suppression test	4.4.4	-	X
Communication equipment test	4.4.5	-	X

NOTE: (1) Without special equipment.
NOTE: (2) Preproduction sample vehicle (Only).

4.5.1.1 Eight mile road test. - Each vehicle shall be operated for a distance of 8 miles or more. For this test vehicles shall be completely assembled and serviced, but need not be combat loaded. Operation shall be on smooth, level, hard-surfaced roads; except that the road shall include a section of approved undulating (bump) road intended to test functioning of suspension and shock absorbers. During this 8 mile test vehicle shall be subjected to all the tests specified in table III at place of manufacture. After the tests, vehicle shall be thoroughly examined for evidence of fuel, and lubricant leakage, or other deficiencies.

4.5.1.2 Speed test. - To determine conformance to 3.5.1, each vehicle shall be driven at minimum, and maximum sustained speeds.

4.5.1.3 Acceleration test. - To determine conformance to 3.5.2, each vehicle shall be given an acceleration test and checked for time required to attain specified speed.

4.5.1.4 Climbing test. - To determine conformance to 3.5.3, vehicle shall be driven up an approved 60 percent grade in forward and reverse gear and vehicle performance observed and recorded.

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4.5.1.5 Engine starting test. - To determine conformance to 3.5.4, vehicle shall be driven up a 60 percent grade, engine stopped for not less than 1 minute then restarted. Vehicle shall be operated on side slopes as specified, engine stopped for not less than 1 minute then restarted.

4.5.1.6 Stopping test. - To determine conformance to 3.5.5.1, the results of 3 stopping tests shall be averaged to make this determination.

4.5.1.7 Holding test. - To determine conformance to 3.5.5.2, the parking brakes shall hold vehicle stationary with engine stopped for not less than 1 minute when parked on an approved 60 percent grade.

4.5.1.8 Turning test. - To determine conformance to 3.5.6, vehicle shall be operated and turned to right and to the left from a pivot to infinite.

4.5.1.9 Fording test. - To determine conformance to 3.5.7, vehicle shall be driven in water of specified depth as indicated in table III and for the specified time.

4.5.1.10 Trench crossing test. - To determine conformance to 3.5.8, vehicles selected as specified in 4.6.1 shall be operated over trenches of specified width and depth.

4.5.1.11 Vertical-obstacle test. - To determine conformance to 3.5.9, vehicle shall be driven over vertical obstacles of specified height.

4.5.1.12 Power train test. - To determine conformance to 3.5.10, vehicle shall be operated throughout all gear and speed ranges, and the brakes applied and released and checked for chatter and slippage.

4.5.1.13 Cooling system test. - To determine conformance to 3.5.11.1 and 3.5.11.2, 1 or more vehicles selected from the first 10 submitted shall be tested in specified ambient temperatures and under all combat operating conditions.

4.5.1.14 Boom and hook test. - To determine conformance to 3.5.12, the boom and hook shall be operated throughout all stages of operation.

4.5.1.15 Boom and hook capacity test. - To determine conformance to 3.5.12.1, the boom and hook shall be operated in the fully extended and retracted positions and lifting specified loads when fully elevated.

4.5.1.16 Winch test. - To determine conformance to 3.5.12.2, 3.5.12.3, and 3.5.12.4, winches, with specified load attached to cables, shall be operated through all ranges of winching operations.

4.5.1.17 Winch brake test. - To determine conformance to 3.5.12.5, when winches are operating under specified loads the brakes shall be operated and checked for holding and clearance when released.

4.5.1.18 Auxiliary hydraulic system power unit test. - To determine conformance to 3.5.12.6, the hydraulic system power unit shall be operated throughout all stages of operation.

4.5.2 Winterization equipment. - To determine conformance to 3.3.3, vehicle shall have the winterization equipment installed and checked for completeness and function of equipment, and vehicle operation.

4.5.3 Stowed equipment. - To determine conformance to 3.3.17, vehicle shall be combat loaded with all on vehicle equipment stowed in their applicable spaces and vehicle checked for completeness, functioning, stowed items, and for interference with any other component.

4.5.4 Radio suppression test. - To determine conformance to 3.5.13, vehicle shall be subjected to a radio suppression test.

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4.5.5 Communication equipment test. - To determine conformance to 3.3.15, vehicle, with the communication equipment installed, the equipment shall be operated throughout all stages of operation, and checked for defects that may cause failure.

4.6 Test failure. - If a vehicle fails to pass any tests specified, the Government inspector shall stop acceptance of subsequent lots until objective evidence has been provided by the contractor that corrective action has been taken.

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery and storage of vehicle shall be in accordance with Specification MIL-T-45056.

6. NOTES

6.1 Intended use. - Vehicles covered by this specification are intended primarily for recovering tanks and other mechanized equipment, especially from the field of battle under fire, by towing or lifting.

6.2 Ordering data. - Contracts or orders should specify the following:

(a) Title, number, and date of this specification.
(b) Preproduction model if required (see 3.1).
(c) Winterization equipment and installation if required (see 3.3.3).

6.3 Color chips. - Until such time as the color chips of FEDERAL STD NO. 595 are changed, color number 2430 of Specification TT-C-595 should be used in lieu of color number 24087 (see 3.6.1).

6.4 Safety precautions. - Caution should be exercised in handling carbon dioxide fire-extinguisher cylinders. They should not be chopped, permitted to strike each other, or be handled roughly, or exposed to solar rays for long periods. Extreme care should be exercised during reinstallation to avoid tripping fire-extinguisher or control system since physical injury is highly probable.

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